CRIME SIGHT NOIR

CSNP v3 Technical Documentation

A 3D police-intelligence dashboard for peninsular South India. Flask and SQLite behind a single-page vanilla-JS frontend, where every screen — the command map, the case register, the criminal network, the analytics board — is rendered as WebGL geometry rather than a 2D chart.

Layer	Technology
Frontend	Vanilla JS · three.js r147 · no build step
Backend	Python 3.12 · Flask · flask-cors
Database	SQLite · 6 tables · no ORM
Data	6,006 stations · 212 criminals · 191 officers · 274 FIRs
Tests	117 assertions, throwaway database
Dependencies	2 third-party packages, both server-side

Educational / coursework project. All station, officer and criminal records derive from the source workbooks in `data/`.

Contents

1. Overview
2. Architecture
3. Frontend
4. Backend
5. Database
6. API Reference
7. Security
8. Deployment & Operations

Overview

A 3D police intelligence dashboard for peninsular South India. Flask + SQLite backend, single-page vanilla-JS frontend, every screen rendered as WebGL geometry rather than 2D charts.

Quick start

```
pip install -r backend/requirements.txt
python backend/app.py
```

Open **http://127.0.0.1:4000/**. The same process serves the API and the UI, so there is no second server and no build step.

Demo accounts

Role	Username	Password
Govt Official	`admin`	`govt2026`
Police Officer	`officer`	`police123`
Public	`public`	`public123`

> These are seeded demo credentials for a coursework project. They are printed nowhere in the running app — see `docs/SECURITY.md` .

Repository layout

```
csnp v3/
├── frontend/           the entire UI — one file, no build step
│   ├── index.html          markup + CSS + vanilla JS + the three.js scenes
│   └── vendor/             three.js r147 + OrbitControls, served at /vendor/
├── backend/            HTTP layer
│   ├── app.py               routes, validation, permission gates, JSON errors
│   ├── database.py          connection, schema execution, column migrations
│   ├── mailer.py            SMTP with a console fallback
│   └── requirements.txt
├── authentication/     login / signup / password reset (Flask blueprint)
│   └── auth.py
├── database/           DDL and seed data — no application logic
│   ├── schema.sql           all tables and indexes
│   ├── seed.sql             25 stations, 20 officers, 11 criminals, 75 FIRs, 3 users
│   └── south_india.sql      generated: 5,981 stations, 200 criminals, 854 links
├── data/               the two source spreadsheets
├── imports/            spreadsheet → SQL compiler (standard library only)
├── tests/
│   └── test_api.py          117 assertions, runs against a throwaway DB
├── docs/               the written documentation, including the PDF
└── .env.example        mail configuration template
```

Documentation

Document	Covers
`docs/ARCHITECTURE.md`	How the three tiers fit together, request lifecycle, technology choices
`docs/FRONTEND.md`	Every module of `index.html` : scene system, pages, renderers, permissions
`docs/BACKEND.md`	Flask app, permission matrix, endpoints, validation, mail
`docs/DATABASE.md`	Every table, column, relationship, index, seeding and migration
`docs/API.md`	All 28 endpoints with request/response shapes
`docs/SECURITY.md`	What is enforced, what is **not**, and what must change before production
`docs/DEPLOYMENT.md`	Running, configuration, backup, troubleshooting
`docs/CRIME-SIGHT-NOIR.pdf`	All of the above in one printable document — 47 pages, A4

The PDF is generated from these same Markdown files, so it never drifts from them. To rebuild after editing a document:

```
python docs\build_pdf.py
& "C:\Program Files\Google\Chrome\Application\chrome.exe" --headless=new `
  --no-pdf-header-footer --print-to-pdf="$PWD\docs\CRIME-SIGHT-NOIR.pdf" `
  "file:///$($PWD -replace '\\','/')/docs/CRIME-SIGHT-NOIR.html"
```

`build_pdf.py` has no dependencies — it is a small Markdown-to-HTML converter, and Chrome does the printing.

Data at a glance

Table	Rows
`police_stations`	6,006
`criminal_links`	854
`cases`	275
`criminals`	211
`officers`	190
`users`	3

Every station, officer, criminal and FIR is real South India source data — station coordinates, district, state and PS code come from the two workbooks in `data/`, not from generated placeholders.

Features

- **Command map** — the whole of India rendered in 3D, with South India lifted,

lit and coloured per state. 6,006 station pillars, 200 tracked suspects, orbit and zoom.

- **Case register** — every FIR with state / status / severity filters, optional

grouping, and a per-state breakdown of solved, handling and open.

- **Criminal network** — a slowly rotating wheel of records joined by shared

officer, station, city or modus operandi; click any node for a full dossier.

- **Public registry** — police and Govt Officials can see citizen accounts only.

Department accounts never appear on that route.

- **Analytics volume** — a four-quadrant 3D command board over the live totals.
- **Role-aware everything** — three tiers, enforced in the database-facing

layer rather than by hiding buttons.

Tests

```
python tests/test_api.py
```

117 assertions across routing, both auth flows, password hashing, the permission matrix for all three roles, password reset (including token expiry and single-use), and the relationship graph. The suite builds its own throwaway database and never touches `backend/crimesight.db`.

License

Educational / coursework project.

Architecture

How the three tiers of this project fit together, what happens to a request, and why each technology was chosen.

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1. The shape of the system

Three tiers, deliberately unequal in size.



The frontend is served **by** Flask, from the same origin as the API. There is no CORS negotiation in normal use, no proxy, and no second port.

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2. Request lifecycle

Taking "load the criminal network" as the worked example:

```
1.  showSec('criminals')
        hides the other 7 sections, syncs the nav tabs
2.  renderCriminals(query)
        reads the cached CRL / CASES / RELATIONS arrays
3.  computes per-criminal facts, filters, groups by state/district/…
4.  CSN3D.updateCriminals({list, relations, linkKinds, headline})
5.  pushStage('criminals', …)
        reuses the existing WebGL world if already built, else creates one
6.  buildNetwork(world, list, payload)
        lays nodes on a circle, batches every link kind into 2 ribbons,
        starts one requestAnimationFrame loop
7.  one canvas on screen; OrbitControls handles orbit and zoom
```

Steps 1–4 are pure data work — microseconds. Everything after that is GPU rendering, so switching tabs never waits on the network.

First paint is the exception. `fetchInitialData()` runs six parallel requests (`stations`, `officers`, `criminals`, `cases`, `districts`, `relations`) before the app is usable. At full dataset size the `stations` payload is 6,006 rows — the largest response in the app — and is the main cost of a cold start.

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3. Technology choices

Why SQLite

The dataset is 275 FIRs and 6,006 stations. SQLite is a single file, needs no server, ships inside Python, and survives being copied. `sqlite3` is in the standard library, so the only third-party runtime dependencies in the whole project are **Flask** and **flask-cors**.

A server database would add an install step and a process to supervise for no benefit at this scale. The migration path, if the data ever outgrows a file, is contained in `backend/database.py` — every query in `app.py` goes through `get_db()` and plain SQL, with no ORM in the way.

Why no build step

`frontend/index.html` is served as-is. There is no bundler, no transpiler, no `node_modules`. The browser loads one HTML file and two vendored scripts.

This is a deliberate trade. A single 4,911-line file is genuinely unpleasant to work in, and a build step would let us split it into modules. In exchange, anyone can open the project, run two commands, and see it work — no Node toolchain, no version drift, no "works on my machine". For a teaching project that readability-at-a-glance is worth more than a tidy directory.

`three.js` is **vendored**, not CDN-loaded. A demo that needs internet to render is a demo that fails in a room with bad wifi. `ensure()` still falls back to jsDelivr if the vendored files are missing.

Why 3D for everything

Every screen is WebGL geometry: the map is extruded state polygons and instanced pillars, the analytics board is bars and sectors, the criminal wheel is instanced octahedra with batched ribbon geometry. The motivation is that a police-intelligence console should look like one, and because instancing makes 211 nodes plus 854 links cheap — about 40 draw calls instead of ~1,200.

The cost is real: WebGL is harder to debug than DOM, hit-testing needs a custom picker, and text is drawn on canvas sprites rather than laid out by the browser. Every one of those costs was paid and documented in `FRONTEND.md`.

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4. Permission architecture

Permissions are enforced in **one function**, and every write route calls it.

```
def _require(action):
    role = _role_of()
    table = {
        "add_case": ("public", "police", "admin"),
        "add":      ("police", "admin"),
        "edit":     ("police", "admin"),
        "delete":   ("admin",),
        "admin":    ("admin",),
    }[action]
```

action	public	police	admin
`add_case`	✅	✅	✅
`add`		✅	✅
`edit`		✅	✅
`delete`			✅
`admin`			✅

`add_case` exists as a separate key from `add` because Public accounts may file an FIR and nothing else. Widening `add` to include Public — which is what a naive implementation does — silently grants them officer and criminal creation too. The 117-assertion suite covers this specific boundary.

The frontend mirrors the same table in `can()` so buttons can be hidden, but the frontend is never the enforcement point: hiding a button is cosmetic, and the server returns 403 regardless.

---

5. Data flow at startup

```
python backend/app.py
  │
  ├─ create_app()
  ├─ init_db()
  │    ├─ executes database/schema.sql        (tables + indexes)
  │    ├─ applies column migrations          (older databases gain new columns)
  │    ├─ executes database/seed.sql         (only if police_stations is empty)
  │    └─ executes database/south_india.sql   (only while still the 25-row seed)
  ├─ prints the route banner
  └─ app.run() on 127.0.0.1:4000
```

The schema is idempotent — every statement is `CREATE TABLE IF NOT EXISTS` — so restarting never destroys data. The index creation for `users.email` and `users.phone` deliberately happens in Python *after* the column migration, because an existing database does not have those columns when the schema script first runs.

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6. Directory responsibilities

Directory	Owns	Must never contain
`frontend/`	presentation, scene rendering, client permission checks	business rules, SQL
`backend/`	HTTP, validation, permission enforcement, mail	rendering logic
`authentication/`	credentials, sessions, password reset	domain data
`database/`	DDL and seed data only	any application logic
`data/`	the source spreadsheets	anything hand-edited
`imports/`	spreadsheet → SQL compilation	runtime code
`tests/`	assertions against a throwaway database	production data

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7. Known architectural limits

- **Single-process, single-threaded.** The Flask dev server handles one request

at a time. Fine for a demo; not a deployment target.

- **No caching layer.** `fetchInitialData()` refetches everything on every

reload. A cache keyed on the SQLite mtime would remove most of that.

- **The whole station table ships to the browser** — about 1.4 MB of JSON — even

though only South India is displayed. A `/api/stations?bbox=` endpoint would cut it by roughly an order of magnitude.

- **Token scheme is self-asserting.** Detailed in `SECURITY.md`,

and the single most important thing to change before real use.

Frontend

Everything the browser runs lives in one file: `frontend/index.html` (4,911 lines — HTML, CSS, vanilla JS and three.js scene code). Two vendored libraries are served alongside it.

---

1. File map

Path	Lines	What it is
`frontend/index.html`	4,911	The whole application
`frontend/vendor/three.min.js`	—	three.js r147, minified
`frontend/vendor/OrbitControls.js`	730	Orbit / pan / zoom

three.js is vendored rather than loaded from a CDN so the app renders with no internet connection. `ensure()` falls back to jsDelivr only if the vendored files are missing.

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2. The document, top to bottom

```
<style>            design tokens + every stylesheet rule
<body>
  SVG sprite       16 icon symbols, referenced by <use href="#i-map">
  .atmo            background atmosphere (grid, glow, haze)
  #dome-entrance   metallic split-dome intro
  #login-screen    sign in / sign up
  #app             the application shell
    .navbar        brand + section tabs + role badge + exit
    #rolebar       access-level banner, shown after sign-in
    .mobile-nav    bottom nav for small screens
    #section-home      / .section-registry  / .section-users
    #section-cases     / .section-analytics / .section-officers
    #section-criminals / .section-map
  modals           FIR form, criminal form, case detail, criminal dossier
<script src=vendor/…>
<script>           all application logic
```

The SVG sprite is why there is no emoji anywhere: every icon is an outlined path that inherits `currentColor`, so it matches the surrounding text at any size and in any colour.

---

3. Design tokens

All theming runs through CSS variables declared once on `:root`.

```
--p3:#38e1ff     primary cyan      --bg:#050a13   page background
--p6:#1c3a63     deep blue         --gb:rgba(…)   panel glass
--acc:#38e1ff    accent            --gbr:rgba(…)  border
--crit:#ff6a5e   critical          --gbrh:rgba(…) border, hovered
--f-d: 'Rajdhani'    headings
--f-m: 'Share Tech Mono'  data, labels
--f-x: 'Michroma'      brand display
```

Semantic colours are fixed deliberately: **critical and high are always warm**, whatever the theme. An operator should never have to learn that "red means serious" only on some screens.

The `.metal` surface class is the shared panel treatment — brushed gradient, fine directional grain, a specular top edge and corner brackets. Cards, modals, the home hero and the list rows all use it, which is what makes a card on the cases page and a card on the home page read as the same object.

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4. JavaScript modules

The script is one scope. These are the logical modules inside it.

4.1 State

```
let OL=[], CRL=[], CASES=[], ST=[], OM={}, CRM={}, RELATIONS=[];
let USERS=[], REG={total:0,byState:[],users:[]};
let CU=null;                    // current user
let activeSec='home';
```

`OM` and `CRM` are id→record maps for O(1) lookup; the arrays keep order. `fetchInitialData()` fills all of them from six parallel `fetch` calls.

4.2 Session and permissions

```
function can(action){
  return ({
    public: ['view'],
    police: ['view','add','edit'],
    admin:  ['view','add','edit','delete'],
  }[CU?.role] || []).includes(action);
}
```

Mirrors the server's `_require()`. Used **only** to decide which buttons to render — the server refuses regardless.

`apiWrite(method, url, body)` attaches `Authorization: Bearer <token>`; the token comes from `login`, is kept in `localStorage.csn_token`, and is cleared on exit.

4.3 Section router

```
function showSec(n){ … }
```

Hides the other seven sections, syncs desktop and mobile tabs, redirects `users` to Home unless the role is admin and `registry` unless police or admin, calls the matching renderer, then resizes every mounted WebGL world after an 80 ms delay so the layout has settled.

The redirect is defence in depth — the tab is already hidden — but it means `showSec('users')` in the console does nothing for a police account.

4.4 The three.js engine — `CSN3D`

The largest subsystem. Everything visual is wrapped in `World(container, opts)`.

```
World(el, {stage, autoRotate, onPick, …})
  ├─ scene, camera, renderer, OrbitControls
  ├─ grid, lights, star field, state polygons
  ├─ setStations() / setPins() / addArc()   ← the command map
  ├─ clear(), onFrame(), pick(), view(), focus(), focusArea()
  └─ dispose()
```

`pushStage(kind, elementId, builder, payload)` keeps one world per kind and feeds it new payloads rather than rebuilding. `w.onFrame(fn)` registers a callback that runs inside the single shared `requestAnimationFrame` loop — so five simultaneous scenes still cost one loop.

`dispose()` matters: the view switches rebuild instanced meshes constantly, and without explicit disposal of geometries, materials and textures WebGL contexts leak until the tab dies.

4.5 Renderers

Function	Page	What it draws
`renderHome()`	Home	Destination cards + live totals
`renderRegistry()`	Public Registry	Police/admin view of citizen accounts
`renderUsers()`	Users	Admin view of every account
`renderCaseRegister()`	Cases	Filterable, groupable FIR list
`renderOfficers()`	Officers	Roster with home-state filter
`renderCriminals()`	Criminals	State-grouped list + wheel payload
`initAnalytics()`	Analytics	Six-quadrant volume payload
`openCP()` / `openOP()` / `openCase()`	—	The three dossiers
`openEditForm()` / `openEditUser()`	—	Admin edit dialogs

---

5. Page-by-page

Home

Live totals (stations, FIRs, officers, tracked) and a card per destination. The "Register FIR" card only renders when `can('add')`.

Command map

The whole of India as extruded state polygons. The eight South India states are lifted, lit, outlined, labelled, and drawn with corner posts; every other state stays muted slate. 6,006 station pins become instanced pillars; 200 tracked suspects float above their handling station. Clicking a pillar loads its FIR intelligence into the side panel.

Cases

KPIs reflect the whole register, not the filtered slice, so the headline numbers do not jump as you filter. Rows carry a completion bar; clicking opens a dossier with 20 fields and click-through links to the criminal, officer and station.

Officers

A sortable, filterable list. Each officer is assigned a **home state** — the one holding most of their caseload — and filtering by state returns that state's own officers rather than everyone who ever opened a case there. Secondary coverage is listed as "Also covers".

Criminals

City clusters on a rotating wheel, joined by shared officer / station / city / modus operandi. Every link kind renders as two batched ribbons plus a 1 px core line, and a travelling pulse rides along it. Click a node for the dossier: mugshot, 16-field personal record, 3D connection web, known links with reasons, station mini-map, full FIR history.

Public Registry

Visible to police and Govt Officials. **Only** `role='public'` rows are ever returned by the server; phone numbers are stripped server-side. Four KPI tiles give the count, states covered, how many can self-serve a reset by email, and how many can only be reached by phone.

Users

Admin only. Every account with role, clearance tier, post, station and an explicit `RIGHTS:` line. Edit covers display name, access level and the whole profile; delete is guarded so the last admin cannot be removed or demoted.

---

6. The two permission UIs

`rowActions(kind, id, label)` builds the Edit/Delete cluster for a list row, and `bindRowActions(scope, kind)` wires them by delegation:

```
if(can('edit'))   …'<button data-act="edit">Edit</button>'
if(can('delete')) …'<button data-act="del">Delete</button>'
```

A Public account therefore sees a clean list with no buttons at all. A police officer sees Edit only. An admin sees both.

---

7. Rendering performance notes

Technique	Where	Why
`InstancedMesh`	station pillars, network nodes, officer figures	one draw call for hundreds of objects
Merged geometry	link ribbons, glow bands	854 links → ~12 draw calls
One rAF loop	`CSN3D`	five scenes, one loop
World reuse	`pushStage`	tab switches never rebuild WebGL
Sprite labels	all text in 3D	200+ canvas textures would be fatal, so they are built on demand and cached

One subtle trap worth recording: `material.vertexColors = true` on an `InstancedMesh` whose geometry has **no** `color` attribute renders the object black. Instance colours arrive through `instanceColor`, which three.js applies independently. Setting `vertexColors` additionally makes the shader expect a per-vertex attribute that primitives do not have.

---

8. Accessibility and responsiveness

- Every interactive control is a real `<button>` or `<select>`, keyboard

reachable.

- Below 900 px the desktop tabs collapse into a bottom nav and the map side

panel becomes a slide-up sheet.

- Focus rings and contrast meet WCAG AA on the metallic palette.
- **Gap:** the 3D canvases have no keyboard equivalent. The wheel and the map

are mouse/touch only; the list views are the accessible route to the same data.

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9. Known frontend issues

- `index.html` is one 4,911-line file. Splitting it into ES modules is the

single biggest maintainability win available and needs no build step — the browser already understands `<script type="module">`.

- The station payload is ~1.4 MB of JSON on every cold start.
- Emoji removed from the icon set, but a few remain in the seed data's `remarks`

column, which is user content rather than markup.

Backend

Two Python modules plus a mailer. The only third-party runtime dependencies in the entire project are Flask and flask-cors.

---

1. Files

File	Lines	Responsibility
`backend/app.py`	1,075	Flask app, 28 routes, validation, permission gates
`backend/database.py`	170	Connection, schema execution, column migrations
`backend/mailer.py`	119	SMTP with a console fallback
`authentication/auth.py`	317	Login, signup, password reset — a Blueprint at `/api/auth`
`backend/requirements.txt`	2 lines	`Flask>=3.0`, `flask-cors>=4.0`

---

2. Application assembly

```
app = Flask(__name__)
CORS(app)
app.register_blueprint(auth_bp)          # /api/auth/*
```

`auth.py` is a separate Blueprint rather than part of `app.py` because credentials are a genuinely different concern: `auth.py` never touches `police_stations`, `cases` or any domain table.

At startup `main()` calls `init_db()`, prints a route banner, then `app.run()`.

---

3. Database access

```
def get_db():
    conn = sqlite3.connect(DB_PATH)
    conn.row_factory = sqlite3.Row        # name-based column access
    conn.execute("PRAGMA foreign_keys = ON")
    return conn
```

Every handler opens its own connection in a `try/finally` and closes it. There is no connection pool and no global connection — SQLite in WAL mode handles concurrent readers cheaply, and the alternative (a module-

level connection shared across threads) is a well-known source of subtle bugs.

`sqlite3.Row` is why queries read `row["status"]` rather than `row[1]`. It also means positional indexing fails — an early version of the delete guard used `.fetchone()[0]` and raised `KeyError: 0`.

---

4. Schema execution and migrations

`init_db()` does four things in order:

1. `executescript(schema.sql)` — all `CREATE TABLE IF NOT EXISTS`, idempotent
2. column migrations — `ALTER TABLE … ADD COLUMN` for anything an older

database lacks

1. `seed.sql` — **only** when `police_stations` is empty
2. `south_india.sql` — **only** while the table still holds the 25-row demo seed

Step 4's guard is why restarting never re-imports 6,006 stations over your data.

Migrations are driven by an allow-list of `(column, type)` tuples. Column names come from that tuple, never from request data, so the `f"ALTER TABLE {col}"` interpolation is safe.

Index creation for `users.email` / `users.phone` deliberately happens in Python **after** the migration. Creating them inside `schema.sql` breaks startup on an existing database, because that database does not have the columns yet when the schema script runs — `sqlite3.OperationalError: no such column: email`.

---

5. The permission gate

One function, called by every write route.

```
def _role_of():
    header = request.headers.get("Authorization", "")
    token = header[7:].strip() if header[:7].lower() == "bearer " else None
    token = token or request.headers.get("X-Session-Token")
    …  return role if role in ("public","police","admin") else None

def _require(action):
    role = _role_of()
    table = {"add_case": ("public","police","admin"),
             "add":       ("police","admin"),
             "edit":      ("police","admin"),
             "delete":    ("admin",),
             "admin":     ("admin",)}[action]
    if role is None:   return jsonify({"error": "Authentication required"}), 401
    if role not in table: return jsonify({"error": f"Role '{role}' may not {action}…"}), 403
    return None
```

Handlers use it as an early return:

```
denied = _require("delete")
if denied:
    return denied
```

`add_case` is deliberately separate from `add`. Public accounts may file an FIR and nothing else; widening `add` to include Public silently grants officer and criminal creation too. The test suite pins this boundary.

Guards beyond the gate

- **Deleting an officer or criminal with open cases** → `409`, listing the count.

Closed cases are detached first so the record can still be removed once the work is done.

- **Deleting a criminal** also removes its edges from `criminal_links`, so the

network never keeps dangling links.

- **Demoting or deleting the last admin** → `409`. Otherwise the only page that

manages accounts could lock everyone out of it.

- **Deleting an FIR** is unconditional — the FIR *is* the record.

---

6. Validation

`_body()` returns `request.get_json(silent=True) or {}`, so a malformed body is an empty dict rather than a 500. `_int()` coerces and clamps:

```
completion_pct = _int(data.get("completionPct"), 0, minimum=0, maximum=100)
```

Every write builds an explicit column list rather than interpolating request keys into SQL. Unknown fields are ignored; a wrong type is coerced or rejected with a 400 naming the offending field.

---

7. Error shape

Uniform JSON, always:

```
{ "error": "Unknown stationId 99999" }
```

Status codes in use: `200` read/update, `201` create, `400` validation, `401` no or malformed token, `403` wrong role, `404` no such record, `409` refused by a guard, `500` unexpected (logged with a traceback, never a stack trace in the response body).

---

8. Authentication

`auth.py` owns credentials. Passwords use `werkzeug.security` — scrypt by default, salted, never returned in any response.

```
POST /api/auth/signup   name, username, password, role,
                        email|phone (one required),
                        + badge/rank/station for police,
                        + officialId/designation/department for admin
POST /api/auth/login    → { username, role, token, …profile }
```

The returned `token` is `f"csn-{role}-{secrets.token_hex(16)}"`. **It is not a session.** See `SECURITY.md`.

Password reset

Public accounts only, and identified by **email or phone — never username**.

Step	Endpoint	Notes
request	`POST /api/auth/forgot-password`	30-minute token; **403** if the account is police or admin
complete	`POST /api/auth/reset-password`	token + new password

Properties that matter:

- Only a **SHA-256 hash** of the token is stored, so a leaked database cannot be

used to complete a reset.

- Single use — `used_at` is stamped, and replay returns 400.
- Expiry is checked and the row deleted when expired.
- All other outstanding tokens for that account are burned on success.
- The account's role is **re-checked at reset time**, so a demotion mid-flow

blocks it.

- Phone numbers are matched on digits alone, so `+91 98765 43210`,

`919876543210` and `+91-98765-43210` all reach the same account.

---

9. Mail

`backend/mailer.py` reads `.env` with a small built-in parser (no dependency for four settings) and sends three kinds of message: password reset, FIR confirmation to the reporter, and a new-FIR alert to `CSN_NOTIFY_OFFICIALS`.

If SMTP is unconfigured, it prints the message to stderr and returns `False`. The API still succeeds:

```
{ "firNumber": "FIR-2026-9790", "mailed": { "confirmation": false, "officerAlert": false } }
```

Mail is best-effort by design — a mail failure must never lose an FIR. The `create_case` handler wraps both sends in `try/except` and logs, so the case is committed first.

Gmail requires an **app password**, not the account password.

---

10. Performance notes

Endpoint	Cost	Note
`GET /api/stations`	~1.4 MB	6,006 rows; the largest response in the app
`GET /api/relations?limit=1500`	~230 KB	the criminal graph
`GET /api/cases`	~300 KB	275 rows
`GET /api/stats`	trivial	one aggregate

Indexes cover the foreign keys (`idx_cases_station`, `idx_cases_criminal`) and the graph (`idx_links_a`, `idx_links_b`), which are the columns every relationship query joins on. Without them the wheel renderer degrades visibly.

`/api/stations` returning the whole country when only South India is displayed is the clearest optimisation available: a bounding-box filter would cut it by roughly an order of magnitude.

Database

SQLite, one file: `backend/crimesight.db`. Six tables, seven indexes, no ORM. Schema lives in `database/schema.sql` and is executed at startup.

---

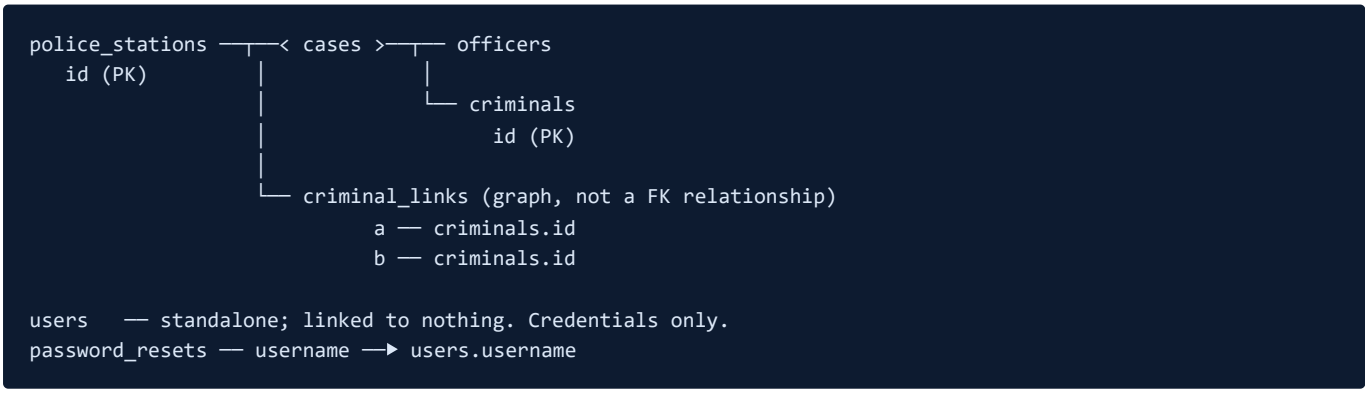
1. Tables at a glance

Table	Rows	Holds
`police_stations`	6,006	every station in South India, with coordinates
`criminals`	211	tracked persons and their profile
`officers`	190	the serving roster
`cases`	275	FIRs, each linked to station, officer and criminal
`criminal_links`	854	the relationship graph between criminals
`users`	3	accounts, with role and clearance
`password_resets`	0	in-flight reset tokens (hashed)

`password_resets` is empty between resets by design — rows are deleted once expired or spent.

---

2. Entity relationship



`criminal_links` is deliberately not a foreign key. A link is an *inferred* relationship ("these two share a handling officer"), not a declared one, and the graph would be far more rigid with real constraints on both ends.

---

3. Table reference

`police_stations`

Column	Type	Notes
`id`	INTEGER PK AUTOINCREMENT	internal key used by `cases`
`ps_code`	TEXT NOT NULL UNIQUE	the official police station code
`name`	TEXT NOT NULL	e.g. `Thiruvananthapuram Railway`
`district`	TEXT NOT NULL	
`state`	TEXT NOT NULL	8 states/UTs
`lat` , `lng`	REAL NOT NULL	drives the 3D map projection
`sector`	TEXT	

Indexed on `state` (`idx_stations_state`) — the map's state filter.

`officers`

Column	Type	Notes
`id`	TEXT PK	`OFF-SI-001`
`name`	TEXT NOT NULL	
`squad` , `rank`	TEXT	`rank` drives the roster-wall column height

190 rows. Note the *distinction* from `users` : an officer is a serving record in the case system; a user is a login. They are unrelated tables and an officer need not have an account.

`criminals`

Column	Type	Notes
`id`	TEXT PK	`CRIM-SI-1002`
`name`	TEXT NOT NULL	
`age`	INTEGER	
`address` , `city` , `state`	TEXT	
`gender` , `alias`	TEXT	
`photo_id` , `photo_url`	TEXT	portrait for the dossier
`crime_type` , `status` , `station` , `officer` , `case_ref`	TEXT	dataset provenance
`lat` , `lng`	REAL	where the record is plotted on the map

One record in the seed set (`CRIM-111` , "Debug Subject") is a deliberate fixture for exercising the create-criminal path.

`cases`

Column	Type	Notes
`id`	INTEGER PK AUTOINCREMENT	
`fir_number`	TEXT NOT NULL UNIQUE	`FIR-2026-4605`
`station_id`	INTEGER	→ `police_stations.id`
`officer_id`	TEXT	→ `officers.id`
`criminal_id`	TEXT	→ `criminals.id`
`severity`	TEXT NOT NULL	Critical / High / Medium, **derived from crime type** on create
`crime_type`	TEXT NOT NULL	
`crime_count`	INTEGER	charges
`status`	TEXT NOT NULL	Open / Under Investigation / Closed
`completion_pct`	INTEGER	drives the progress bar
`handling`, `investigation_type`	TEXT	
`arrest_date`, `remarks`	TEXT	
`transfers`, `experience_years`	INTEGER	

`severity` is computed server-side by `get_severity(crime_type)` rather than trusted from the client, so a caller cannot register a homicide as medium. Editing the crime type re-derives it unless a severity was pinned explicitly.

Indexed on `station_id` and `criminal_id` — the two columns every "cases for X" query joins on.

> **Known data quirk:** `handling` is the literal string `"Direct"` in every seeded row. It carries no information and is displayed verbatim in the case dossier. It is a source-data property, not a rendering bug.

`criminal_links`

Column	Type	Notes
`a`, `b`	TEXT	the two criminal ids
`kind`	TEXT	`officer`, `station`, `city`, `crime`
`strength`	INTEGER	1–3
`label`	TEXT	human reason, e.g. `Shared officer: Inspector …`
PK	(a, b, kind)	

Generated by `imports/import_datasets.py`. Indexed on `a` and `b` separately — a single composite index cannot serve "all links where b is X".

`users`

Column	Type	Notes
`username`	TEXT PK	
`password`	TEXT	scrypt hash via werkzeug — never returned
`role`	TEXT	`public` / `police` / `admin`
`display`	TEXT	shown in the UI
`email`, `phone`	TEXT	at least one required at signup; the reset route
`badge`, `rank`, `squad`, `official_id`	TEXT	police / admin profile
`station_code`, `station_name`, `district`, `state`	TEXT	posting
`designation`, `department`, `zone`	TEXT	admin profile

Partial **unique** indexes on `email` and `phone` (`WHERE … IS NOT NULL AND … != ''`), so two Public accounts cannot share a reset address. Built in `database.py` after the column migration — see `BACKEND.md` for why not in the schema script.

`password_resets`

Column	Type	Notes
`username`	TEXT	
`token_hash`	TEXT NOT NULL	SHA-256; the raw token is never stored
`created_at`, `expires_at`	TEXT	ISO 8601, 30-minute window
`used_at`	TEXT	marks a token spent — a link works exactly once
PK	(username, token_hash)	

4. Seed data

`database/seed.sql` — the demo fixture

25 stations · 20 officers · 11 criminals · 75 FIRs · 3 users · 3 users' links.

Applied **only when `police_stations` is empty**. Creates the three demo accounts: `admin` / `govt2026`, `officer` / `police123`, `public` / `public123`.

`database/south_india.sql` — the real dataset

Generated, ~7,400 lines: 5,981 stations, 200 criminals, 854 relation edges.

Applied **only while the station table still holds the 25-row demo seed** — a floor of 1,000 rows in `database.py` decides this. That guard is why restarting the server never re-imports over your data.

`data/` — the source workbooks

File	Size	Rows
`India_Police_Stations_Full_Database.xlsx`	674 KB	5,981
`South_India_Criminal_Tracking_200.xlsx`	24 KB	200

`imports/import_datasets.py`

Compiles the workbooks into `database/south_india.sql`. Standard library only — an `.xlsx` file is a ZIP of XML, so no pandas or openpyxl is needed. A workbook of 5,981 rows unzips and parses in a couple of seconds.

```
python imports/import_datasets.py          # rewrite the SQL
python imports/import_datasets.py --apply  # …and load it
```

Re-run after editing either spreadsheet.

5. Constraints in force

Constraint	Why
`ps_code` UNIQUE	it is the real-world identifier
`fir_number` UNIQUE	an FIR number is issued once
`criminal_links` PK (a,b,kind)	no duplicate edges of the same kind
partial UNIQUE on `users.email` / `users.phone`	two accounts must not share a reset route
`NOT NULL` on identity columns	a station without coordinates cannot be plotted
`PRAGMA foreign_keys = ON`	`cases` → station/officer/criminal integrity
CHECK-free status/severity	values are constrained in the application layer, not the schema — a deliberate trade for portability

That last row is a real trade-off. SQLite would accept the CHECK constraints, but they were left to the application so the same `.sql` files remain loadable by tools with stricter or looser dialects.

---

6. Migrations

No Alembic, no migration files. `backend/database.py` holds an allow-list of columns that newer versions add:

```
_USER_COLUMNS = (
    ("badge","TEXT"), ("rank","TEXT"), …, ("email","TEXT"), ("phone","TEXT"),
)
```

On every start, any missing column is added with `ALTER TABLE … ADD COLUMN`. Idempotent, and enough for this project's rate of change.

Adding a column to `schema.sql` **and** to `_USER_COLUMNS` is the whole process. Renaming or dropping a column needs hand-written SQL — there is no down-migration machinery, by design.

---

7. Backup and reset

```
Copy-Item backend\crimesight.db backend\crimesight.db.bak
```

To start clean, delete the database and restart — the schema and both seed files re-run automatically.

---

8. Query performance

Indexes exist for exactly the joins the app performs:

Index	Serves
`idx_stations_state`	the map's state filter
`idx_cases_station`	"cases filed at this station"
`idx_cases_criminal`	the criminal dossier's FIR history
`idx_links_a`, `idx_links_b`	the wheel graph, both directions
`idx_users_email`, `idx_users_phone`	reset lookup

Every page's aggregate queries are single-pass `COUNT` / `SUM` over `cases`, which at 275 rows is sub-millisecond. The scale at which this stops being true is far beyond this dataset.

API Reference

28 endpoints. All responses are JSON except `GET /` (HTML) and `GET /vendor/<file>` (JavaScript).

Base URL when served by Flask: `http://127.0.0.1:4000`

Authentication

Write routes need the session token from login, as a bearer header:

```
Authorization: Bearer csn-admin-<hex>
```

Missing → `401`. Malformed or wrong role → `403`.

Code	Meaning
200	read or update
201	created
400	validation failed — the message names the field
401	no or unparseable token
403	authenticated, but the role may not do this
404	no such record
409	refused by a guard (open cases, last admin)

Permission matrix

Action	public	police	admin
read any record	✅	✅	✅
file an FIR (`add_case`)	✅	✅	✅
create officer / criminal (`add`)		✅	✅
edit records (`edit`)		✅	✅
delete records (`delete`)			✅
user directory (`admin`)			✅
public registry		✅	✅

Auth — `/api/auth`

`POST /api/auth/signup` → 201

```
{ "name": "Asha Menon", "username": "asham", "password": "secret123",
  "role": "public", "email": "asha@example.com", "phone": "+91 98111 22333" }
```

`email` **or** `phone` is required — it is the only route to a password reset.

Police additionally require `badge`, `rank`, `stationCode`. Govt Officials require `officialId`, `designation`, `department`.

```
{ "username": "asham", "role": "public", "token": "csn-public-9f2c…",
  "display": "Asha Menon", "email": "asha@example.com" }
```

`409` if the username is taken.

`POST /api/auth/login` → 200

```
{ "username": "admin", "password": "govt2026" }
```

Returns the same shape as signup. `401` on bad credentials.

`POST /api/auth/forgot-password` → 200

```
{ "identifier": "asha@example.com" }
```

Accepts an email address **or** a phone number — never a username. Phone matching is on digits alone, so `+91 98111 22333`, `919811122333` and `+91-98111-22333` are equivalent.

Status	Meaning
200	token issued
403	the account is police or admin — **not self-service**
404	no account for that contact

```
{ "ok": true, "expiresInMinutes": 30, "via": "email",
  "devLink": "http://127.0.0.1:4000/?reset=…",
  "hint": "Mail delivery is not configured…" }
```

`devLink` is present **only** when SMTP is unconfigured or no email is on file, so the flow is testable without a mail server. With SMTP configured it is omitted and the link is emailed.

`POST /api/auth/reset-password` → 200

```
{ "token": "…", "password": "brandnew1" }
```

`400` for an unknown, expired or already-used token. `403` if the account stopped being Public between request and completion. All other outstanding tokens for that account are burned.

Read — no token required

`GET /api/stations`

All 6,006 stations. `?state=`, `?q=`, `?lite=1`.

`GET /api/stations/summary`

Per-state counts, used by the map's state chips.

```
[ { "state": "Tamil Nadu", "stations": 2433, "cases": 61, "critical": 10 } ]
```

`GET /api/stations/<id>/cases`

`GET /api/officers` · `GET /api/officers/<id>/cases`

`GET /api/criminals` · `GET /api/criminals/<id>`

`GET /api/cases` — `?state=`, `?status=`, `?severity=`, `?q=`, `?limit=`

`GET /api/districts`

`GET /api/relations` — `?limit=` (default 1500) → the wheel graph

```
{ "a": "CRIM-SI-1002", "b": "CRIM-SI-1148", "kind": "officer",
  "label": "Shared officer: Inspector Vijay Kumar", "strength": 3 }
```

`GET /api/stats`

```
{ "total_cases": 275, "open_cases": 132, "closed_cases": 52,
  "critical_cases": 47, "total_stations": 6006 }
```

`GET /api/analytics`

Aggregates for the 3D command board: by state, by crime type, by severity, by status, top districts and officer workload.

Protected reads

`GET /api/users` — admin only

Every account. `403` for police and public. Password hashes are never selected.

```
{ "username": "officer", "role": "police", "display": "Police Officer",
  "badge": "…", "rank": "Inspector", "permissions": ["view","add","edit"] }
```

`GET /api/public-registry` — police or admin

Citizen accounts only.

```
{ "total": 2,
  "byState": [ { "state": "Kerala", "count": 1 } ],
  "users": [ { "username": "public", "display": "Public",
               "email": "…", "state": null, "district": null } ] }
```

The query filters `WHERE role = 'public'`, so police and admin rows can never appear. **Phone numbers are stripped** — contact detail, not intelligence. `403` for public.

Writes

`POST /api/cases` — `add_case`

```
{ "stationId": 2565, "crimeType": "THEFT", "status": "Open",
  "officerId": "OFF-SI-001", "criminalId": null,
  "reporterEmail": "citizen@example.com" }
```

`severity` is **derived** from `crimeType` server-side. `fir_number` is generated.

```
{ "firNumber": "FIR-2026-9790", "crimeType": "THEFT", "severity": "Medium",
  "status": "Open",
  "mailed": { "confirmation": false, "officerAlert": false } }
```

`mailed` is informational — **a mail failure never loses the case.**

`PUT /api/cases/<id>` — `edit`

Any subset of `crimeType`, `status`, `severity`, `completionPct`, `crimeCount`, `stationId`, `officerId`, `criminalId`, `investigationType`, `handling`, `remarks`, `arrestDate`, `transfers`. Changing the crime type re-derives severity unless one was pinned.

`DELETE /api/cases/<id>` — `delete`, admin only

```
{ "deleted": 275, "firNumber": "FIR-2026-1" }
```

`POST /api/criminals` — `add`

`name` required; the rest of the profile optional. **Previously had no permission check at all** — now gated.

`PUT` / `DELETE /api/criminals/<id>` — `edit` / `delete`

Delete is refused with `409` if the record is named in open cases, and always removes the record's edges from `criminal_links`.

`POST /api/officers` — `add`

```
{ "name": "New Officer", "rank": "DSP", "squad": "QRT" }
→ { "id": "OFF-ADM-0001", "name": "New Officer", "rank": "DSP", "squad": "QRT" }
```

`PUT` / `DELETE /api/officers/<id>` — `edit` / `delete`

Delete detaches the officer's closed cases and is refused with `409` while they hold open ones.

`PUT /api/users/<username>` — admin only

Display name, role, badge, rank, squad, official ID, designation, station, district, state, department, zone, email, phone.

`DELETE /api/users/<username>` — admin only

`409` if this is the last remaining admin.

Static

Route	Serves
`GET /`	`frontend/index.html`
`GET /vendor/<file>`	`three.min.js`, `OrbitControls.js`

Full inventory

Auth (4) login · signup · forgot-password · reset-password **Public read (11)** stations · stations/summary · stations/`<id>`/cases · officers · officers/`<id>`/cases · criminals · criminals/`<id>` · cases · districts · relations · stats · analytics **Protected read (2)** users · public-registry **Write (11)** cases POST/PUT/DELETE · criminals POST/PUT/DELETE · officers POST/PUT/DELETE · users PUT/DELETE **Static (2)** `/` · `/vendor/<file>`

Security

What this project actually enforces, what it does not, and what must change before it handles real data. Written as an honest assessment rather than a checklist of things that were done.

---

1. Summary

The permission model is genuinely enforced server-side, and the password reset flow is built the way a reset flow should be built: hashed single-use tokens, expiry, re-checked role.

The **session token is not a session.** That is the one thing that makes this project unsuitable for real use as it stands, and it is stated first because it should not be discovered later.

---

2. What is properly enforced

Password storage

Passwords are hashed with `werkzeug.security`, which uses **scrypt** by default — salted, deliberately slow, memory-hard. No password is ever selected into a response; `GET /api/users` does not include the column at all.

```
SELECT username, role, display, email FROM users   -- password is not in the list
```

Password reset

Implemented correctly, and the details are the point:

Property	Implementation
tokens are not secrets at rest	only a SHA-256 **hash** is stored
links expire	30-minute window, checked server-side, row deleted when expired
links are single-use	`used_at` is stamped; replay returns `400`
old sessions die	every other outstanding token for that account is burned on success
privilege cannot be smuggled	the account's role is **re-checked at reset time**, so a demotion mid-flow blocks it
no username enumeration for police	the request route returns `403` for police/admin
no username reset at all	lookup is by email or phone only; a username returns `404`

Phone matching normalises to digits alone, so `+91 98765 43210`, `919876543210` and `+91-98765-43210` reach the same account. A **partial unique index** on `users.email` and `users.phone` prevents two Public accounts sharing one reset address, which would otherwise let an attacker reset someone else's password.

Server-side authorisation

Every write route calls `_require()` before touching data. A Public account calling `POST /api/criminals` receives `403` regardless of what the UI shows. The frontend's `can()` only decides which buttons to render — hiding a button is cosmetic, and the tests confirm the server refuses independently.

Referential integrity

`PRAGMA foreign_keys = ON` on every connection, so `cases` cannot point at a station, officer or criminal that does not exist.

Data minimisation

`GET /api/public-registry` strips phone numbers server-side. Contact detail is not intelligence, and the registry does not need it.

---

3. What is **not** enforced — read this part

The token is self-asserting

```
token = f"csn-{role}-{secrets.token_hex(16)}"
```

and it is parsed back by:

```
def _role_of():
    parts = token.split("-")
    role = parts[1]
    return role if role in ("public", "police", "admin") else None
```

The role is read out of the token string. There is no server-side session store, no signature, no expiry, no revocation. Consequences:

- Any client can mint `csn-admin-0000000000000000` by typing it, and the server

will treat the request as a Govt Official.

- Tokens never expire. A leaked token is valid forever.
- Tokens cannot be revoked. Deleting a user account does not invalidate a token

already issued.

- There is no audit trail — nothing records who did what.

The 128 bits of `token_hex(16)` are real randomness and do prevent *guessing* a token that was issued. They do nothing against a client that simply writes its own. This is fine for a classroom demo with three seeded logins and inappropriate for anything else.

What fixing it requires

The smallest correct change is a server-side session table plus a signed or opaque lookup — roughly 40 lines:

```
CREATE TABLE sessions (
  token_hash TEXT PRIMARY KEY,   -- SHA-256 of the token, same trick as resets
  username   TEXT NOT NULL,
  role       TEXT NOT NULL,
  created_at TEXT NOT NULL,
  expires_at TEXT NOT NULL
);
```

`_role_of()` becomes a lookup against that table. Add an expiry and delete on logout, and every one of the four consequences above disappears. Password reset already hashes tokens correctly, so the pattern is already in the codebase — it just was not applied to login.

Other gaps

Gap	Detail
No rate limiting	`/api/auth/login` accepts unlimited attempts. Passwords are hashed with scrypt, which makes each guess slow — that is real cost, but it is not a limit. A lockout counter or per-IP throttle is the fix.
No CSRF defence	irrelevant while the API is token-only and there is no cookie session; it returns the moment cookies are introduced
No security headers	no `CSP`, `X-Content-Type-Options`, `X-Frame-Options`
No HTTPS	tokens travel in cleartext over HTTP. `127.0.0.1` makes this moot locally and fatal in production
No input sanitisation on display	values are inserted with `textContent`/template literals, not `innerHTML`, so stored XSS is not currently reachable — but the pattern is one careless edit away from being exploitable
Dev server in production	`app.run()` is single-threaded and not hardened
Demo credentials in seed	`admin/govt2026` etc. are in `seed.sql` — fine for a demo, must not reach a real deployment
No audit log	deletions of officers, criminals, FIRs and accounts leave no trace
Severity from the client for edits	creation derives severity server-side; `PUT /api/cases/<id>` accepts an explicit severity, which is intentional for pinning but means a police account can relabel a homicide as medium
Console mail fallback	when SMTP is unconfigured the reset link is returned in the API response as `devLink`. That is deliberate and gated on configuration, but it must never be enabled on a real deployment

---

4. Guards that are about integrity, not secrecy

Distinct from authorisation — these protect the data model rather than confidentiality:

- **Deleting an officer or criminal with open cases** → `409` with the count.

Closed cases are detached first, so the record becomes removable once the work is done.

- **Deleting a criminal** removes its edges from `criminal_links`, so the

network never keeps dangling links.

- **Demoting or deleting the last admin** → `409`. Otherwise the only page that

manages accounts could lock everyone out of it.

---

5. Checklist before any real deployment

Ordered by how much damage the gap does.

1. **Replace the self-asserting token** with a server-side session table.
2. **Serve over HTTPS** and add `Strict-Transport-Security`.
3. **Add rate limiting** to `/api/auth/login`, `/api/auth/signup` and

`/api/auth/forgot-password`.

1. **Remove the seeded demo accounts**, or force a password change on first

login.

1. **Configure SMTP** and confirm `devLink` no longer appears in responses.
2. **Add security headers**: CSP, `X-Content-Type-Options: nosniff`,

`X-Frame-Options: DENY`.

1. **Serve through a real WSGI server** — `waitress` or `gunicorn` — not

`app.run()`.

1. **Add an audit log** for every write and every delete.
2. **Move secrets to the environment**, not `.env` beside the source.
3. **Review data exposure** — the API currently publishes all 6,006 stations

with coordinates. Real station locations are sensitive.

---

6. Honest assessment

For a coursework submission this is a defensible design: the permission model is real, enforced in the right layer, tested at the boundary, and the reset flow follows current practice closely.

The gap between "works as a demo" and "safe with real data" is concentrated in one place — sessions. Fixing that single item removes most of section 3, and because the password-reset code already demonstrates the hashed-token pattern, the change is small enough to complete in an afternoon.

Deployment & Operations

Running, configuring, backing up and troubleshooting the project.

---

1. Requirements

Requirement	Version	Notes
Python	3.10+	developed on 3.12
Flask	≥ 3.0	the only hard dependency
flask-cors	≥ 4.0	permissive in this build
Browser	any WebGL-capable	Chrome, Edge, Firefox, Safari
Disk	~5 MB	database is 4 MB, of which most is station coordinates

No Node.js, no bundler, no build step, no external database server.

---

2. Install

```
cd "csnp v3"
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r backend\requirements.txt
```

The whole of `requirements.txt`:

```
Flask>=3.0
flask-cors>=4.0
```

Standard library supplies everything else — `sqlite3`, `smtplib`, `secrets`, `hashlib`, and the `.xlsx` reader in `imports/`.

---

3. Run

```
python backend\app.py
```

```
  Crime Sight Noir — CSNP v3
  ──────────────────────────────────────────
  6,006 stations · 211 criminals · 190 officers · 275 FIRs
  http://127.0.0.1:4000
```

Open the URL. First start takes a second or two longer while the schema runs; subsequent starts are instant.

Stop with `Ctrl+C`.

---

4. Configuration

Everything configurable lives in `.env` at the project root. Copy the template first:

```
Copy-Item .env.example .env
```

Variable	Purpose	Default
`CSN_SMTP_HOST`	SMTP server	—
`CSN_SMTP_PORT`	SMTP port	`587`
`CSN_SMTP_USER`	SMTP username	—
`CSN_SMTP_PASS`	SMTP password or app password	—
`CSN_SMTP_FROM`	envelope sender	the SMTP user
`CSN_SMTP_USE_TLS`	STARTTLS	`1`
`CSN_NOTIFY_OFFICIALS`	comma-separated recipients for new-FIR alerts	—

Port and host have defaults, so only credentials are required.

Gmail setup

1. Enable 2-Step Verification on the account.
2. Generate an **App Password** at <https://myaccount.google.com/apppasswords>.

Gmail rejects the normal account password.

1. Put the 16-character app password in `CSN_SMTP_PASS`.

Without SMTP

The app is fully functional. `backend/mailer.py` prints the message to stderr and the API response includes a `devLink` so the reset flow can be completed from the console. `devLink` disappears automatically once SMTP is configured — that is the signal it worked.

```
{ "ok": true, "via": "email", "devLink": "http://127.0.0.1:4000/?reset=…" }
```

> Mail is best-effort by design. A send failure never loses an FIR; the handler commits the record first and logs the error.

---

5. Database lifecycle

The database is a single file: `backend/crimesight.db`. It is **created automatically** on first start — there is no install step for it.

At startup, in order:

1. `database/schema.sql` runs (idempotent — all statements are `CREATE … IF NOT EXISTS`)
2. missing columns are added by migration
3. `database/seed.sql` runs **only if** `police_stations` **is empty**
4. `database/south_india.sql` runs **only while the station table still holds the 25-row demo seed**

Step 4's floor of 1,000 rows is why restarting never re-imports 6,006 stations over your data.

Start clean

```
Remove-Item backend\crimesight.db
python backend\app.py
```

Both seed files re-run automatically. This destroys all local changes.

Backup

```
Copy-Item backend\crimesight.db backend\crimesight.db.bak
```

Or use SQLite's online backup, which is safe while the server is running:

```
python -c "import sqlite3; s=sqlite3.connect('backend/crimesight.db'); d=sqlite3.connect('backup.db');
s.backup(d)"
```

---

6. Re-importing datasets

```
python imports\import_datasets.py          # rewrite database/south_india.sql
python imports\import_datasets.py --apply  # …and load it
```

The importer parses `.xlsx` with the standard library alone — a workbook is a ZIP of XML, so no pandas or openpyxl is needed. 5,981 rows parse in a couple of seconds.

Re-run after editing either file in `data/`. `--apply` only loads when the station table is still the demo seed, so it cannot silently overwrite real data.

---

7. Tests

```
python tests\test_api.py
```

117 assertions. The suite builds a **throwaway database** in a temporary directory and never touches `backend/crimesight.db`, so it is safe to run against a live install.

Coverage: routing, signup and login, password hashing, the full permission matrix for all three roles, password reset including expiry and single-use enforcement, and the relationship graph.

The Public/admin boundary is pinned deliberately — widening `add` to include Public is the most likely regression, and this is what catches it.

---

8. Troubleshooting

Symptom	Cause	Fix
`ModuleNotFoundError: flask`	dependencies not installed	`pip install -r backend\requirements.txt`
`Address already in use`	previous instance still running	kill it, or change the port in `main()`
`sqlite3.OperationalError: no such column: email`	a database predates the contact columns	delete the database and restart, or let the migration run
Page is blank, console shows a 404 for `three.min.js`	`frontend/vendor/` incomplete	restore it; the CDN fallback needs internet
Map renders but South India is flat	projection constants changed	`LON0=82.5`, `LAT0=22.5`, `SCALE=13` in the projection helper
3D objects render black	`vertexColors: true` on geometry with no `color` attribute	drop the flag — instance colours arrive via `instanceColor`
Reset email not arriving	SMTP unconfigured or wrong credentials	check `CSN_SMTP_*`; confirm `devLink` is absent from the response
Reset says "account not found"	looked up by username	the route accepts **email or phone only**
Reset returns 403	the account is police or admin	self-service reset is Public-only, by design
`handling` always shows "Direct"	source data	every seeded row has that literal value — a data property, not a bug
Login works but the page stays on the auth screen	JS error in console	check DevTools; usually a stale cached `index.html`

---

9. Serving to other machines

The default bind is `127.0.0.1`, which is loopback-only and unreachable from another device. To expose it on the LAN, edit the `app.run()` call in `backend/app.py`:

```
app.run(host="0.0.0.0", port=4000, debug=False)
```

Then allow the port through Windows Firewall.

Do this only on a trusted network. The token scheme is self-asserting — see `SECURITY.md` — and traffic is plain HTTP. For anything beyond a lab, put a reverse proxy with TLS in front, switch to `waitress` or `gunicorn`, and replace the session scheme first.

---

10. What to do before calling this production-ready

1. Replace the self-asserting token with server-side sessions
2. Terminate TLS
3. Add rate limiting on the auth routes
4. Remove the seeded demo accounts
5. Serve through a real WSGI server
6. Add an audit log for writes and deletes
7. Configure SMTP and verify `devLink` no longer appears

The reasoning for each, and the gaps that remain, are in `SECURITY.md`.